Fuzzy Logic and ANFIS Based Intelligent System

1. Introduction

Fuzzy logic is a technique used to model uncertainty and approximate reasoning. It allows systems to make decisions similar to human thinking by using linguistic variables instead of precise numerical values.

ANFIS (Adaptive Neuro-Fuzzy Inference System) combines fuzzy logic with neural networks. It uses learning algorithms to automatically adjust membership functions and improve prediction accuracy.

2. Objective

The objectives of this project are:

- To design a fuzzy logic system for determining washing machine cycle time
- To develop a hybrid intelligent system using ANFIS to predict student performance
- To understand the integration of fuzzy logic and neural networks

3. Q1: Fuzzy Logic System Design (Washing Machine)

3.1 Inputs and Output

Inputs:

- Dirt Level
- Load Size

Output:

- Washing Time

3.2 Fuzzy Sets Definition

Dirt Level:

- Low
- Medium
- High

Load Size:

- Small
- Medium

- Large

Washing Time:

- Short
- Medium
- Long

3.3 Membership Functions

Triangular and trapezoidal membership functions are used.

- Dirt Level range: 0–10
- Load Size range: 0–10
- Time range: 0–60

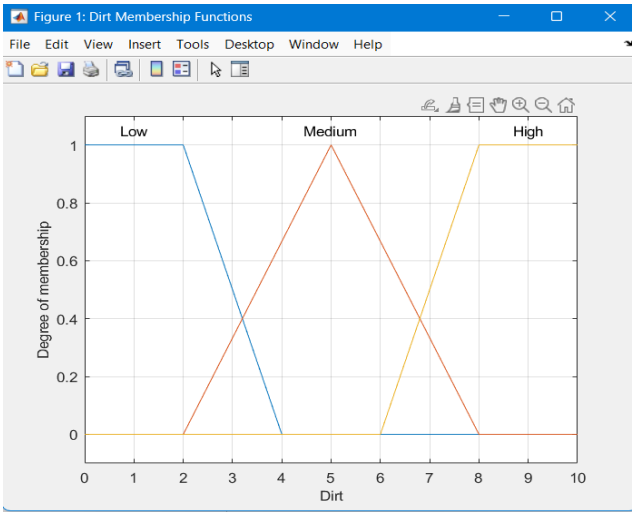

Figure1:Dirt Membership Function

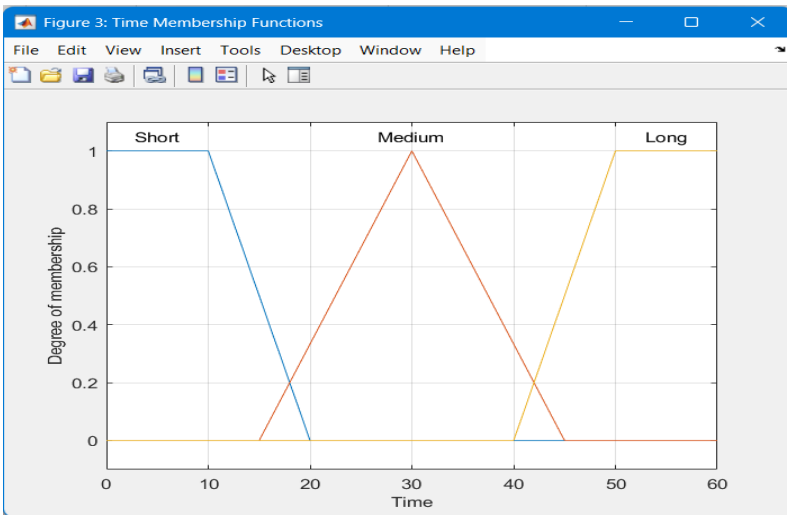

Figure3:Time Membership Function

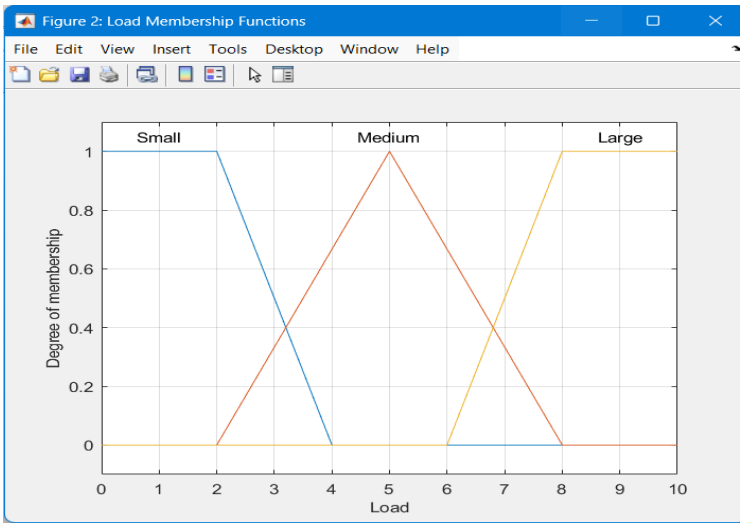

Figure2:Load Membership Function

3.4 Fuzzy Rule Base

1. IF Dirt is Low AND Load is Small → Time is Short
2. IF Dirt is Medium AND Load is Small → Time is Medium
3. IF Dirt is High AND Load is Small → Time is Long
4. IF Dirt is Low AND Load is Medium → Time is Short
5. IF Dirt is Medium AND Load is Medium → Time is Medium
6. IF Dirt is High AND Load is Medium → Time is Long
7. IF Dirt is Low AND Load is Large → Time is Medium
8. IF Dirt is Medium AND Load is Large → Time is Long
9. IF Dirt is High AND Load is Large → Time is Long

3.5 MATLAB Implementation

The system is implemented using MATLAB Fuzzy Logic Toolbox. The rule viewer shows how inputs map to output.

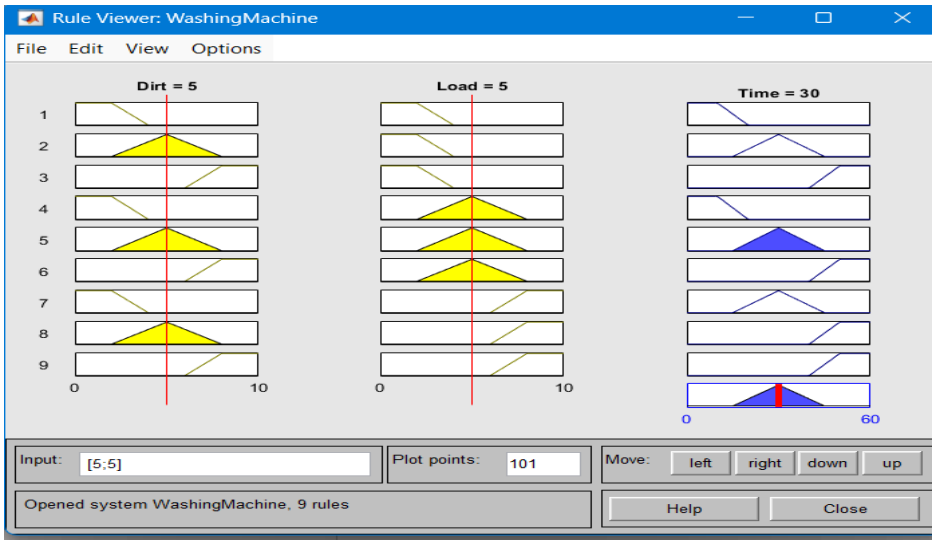

Figure4:Rule Viewer output

3.6 Results

Test outputs:

- Dirt = 2, Load = 2 → Time ≈ 7.62 minutes
- Dirt = 5, Load = 5 → Time ≈ 30 minutes
- Dirt = 8, Load = 9 → Time ≈ 52.38 minutes

```
Command Window

ans =

 9×69 char array

  '1. If (Dirt is Low) and (Load is Small) then (Time is Short) (1)     '
  '2. If (Dirt is Medium) and (Load is Small) then (Time is Medium) (1) '
  '3. If (Dirt is High) and (Load is Small) then (Time is Long) (1)     '
  '4. If (Dirt is Low) and (Load is Medium) then (Time is Short) (1)    '
  '5. If (Dirt is Medium) and (Load is Medium) then (Time is Medium) (1)'
  '6. If (Dirt is High) and (Load is Medium) then (Time is Long) (1)    '
  '7. If (Dirt is Low) and (Load is Large) then (Time is Medium) (1)    '
  '8. If (Dirt is Medium) and (Load is Large) then (Time is Long) (1)   '
  '9. If (Dirt is High) and (Load is Large) then (Time is Long) (1)     '

------------- TEST OUTPUTS -------------
Input (Dirt=2.0, Load=2.0) --> Time = 7.62 minutes
Input (Dirt=5.0, Load=5.0) --> Time = 30.00 minutes
Input (Dirt=8.0, Load=9.0) --> Time = 52.38 minutes
```

Figure5:Output

4. Q2: Hybrid Intelligent System (ANFIS)

4.1 Inputs and Output

Inputs:

- Attendance
- Assignment Marks
- Test Marks

Output:

- Performance Level (Poor, Average, Good)

4.2 Dataset

A dataset of 1000 student records is used.

4.3 Integration of Fuzzy Logic and Neural Networks

Fuzzy logic provides rule-based reasoning. ANFIS uses neural network learning to adjust membership functions and improve prediction accuracy.

4.4 Methodology

- Data normalization applied
- Train-test split (80% training, 20% testing)
- ANFIS training using MATLAB
- Validation used to monitor performance
- Early stopping applied

4.5 Results

- Training RMSE = 0.0311324
- Checking RMSE = 0.0323606
- Best Epoch = 1
- Final Test RMSE = 3.24

The model shows stable performance with minimal difference between training and checking error.

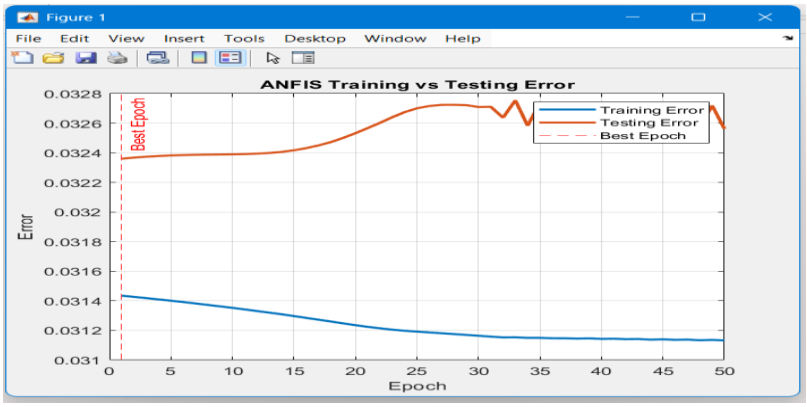

Figure 6:ANFIS Training VS Testing Error

4.6 Output Example

Input:

- Attendance = 85
- Assignment = 80
- Test = 78

Output:

- Predicted Score = 80.09
- Performance Level = Good

```
Minimal training RMSE = 0.0311324
Minimal checking RMSE = 0.0323606
Best Epoch = 1
Final Test RMSE = 3.24

Predicted Score = 80.09
Performance Level = Good
```

Figure 7: Output

4.7 Training Process

The training process includes initialization and iterative error reduction across epochs.

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Command Window

ANFIS info:
        Number of nodes: 34
        Number of linear parameters: 32
        Number of nonlinear parameters: 18
        Total number of parameters: 50
        Number of training data pairs: 800
        Number of checking data pairs: 200
        Number of fuzzy rules: 8

Start training ANFIS ...

1       0.031435        0.0323606
2       0.0314267       0.0323679
3       0.0314183       0.0323738
4       0.0314096       0.0323785
Step size increases to 0.011000 after epoch 5.
5       0.0314009       0.0323822
6       0.031392        0.0323848
```

Figure 8: Training Process

5. Conclusion

The project successfully demonstrates the use of fuzzy logic and ANFIS for intelligent decision-making and prediction.

6. References

- MATLAB Documentation
- Fuzzy Logic Toolbox
- ANFIS Tutorials